

```

%root_path = fileparts(mfilename("fullpath"));
root_path = pwd;
data_path = fullfile(root_path,'sym4D_cutsAndFits');

if ~exist('cuts2fit200GP','var')
    ld = load(fullfile(data_path,'multicuts_fit_dataGH_ei200meV.mat'));
    cuts2fit200 = ld.cuts2fit200;
end
if exist('used_ds200GP','var')
    used_ds = used_ds200GP;
else
    used_ds = containers.Map()
    sel_info = struct('cuts',[],'en_range',[],'fname','','bg_par',[]);
    %
    sel_info.cuts = {cuts2fit200GP.cuts_off110f};
    sel_info.en_range = 30:10:120;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200off110_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br110dir111')=sel_info;
    %
    sel_info.cuts = {cuts2fit200GP.cuts_off200f};
    sel_info.en_range = 30:10:130;
    sel_info.cut_dim = 2;
    sel_info.fname='EnFit_Ei200off200_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir111')=sel_info;
    %
    used_ds200GP = used_ds;
end

```

```
used_ds =
```

```
Map with properties:
```

```

    Count: 0
    KeyType: char
    ValueType: any

```

```

select= 'Ei200Br200dir111';
sel_info = used_ds(select);

cuts_list = sel_info.cuts

```

```

cuts_list = 1x1 cell array
    {1x1 sqw}

```

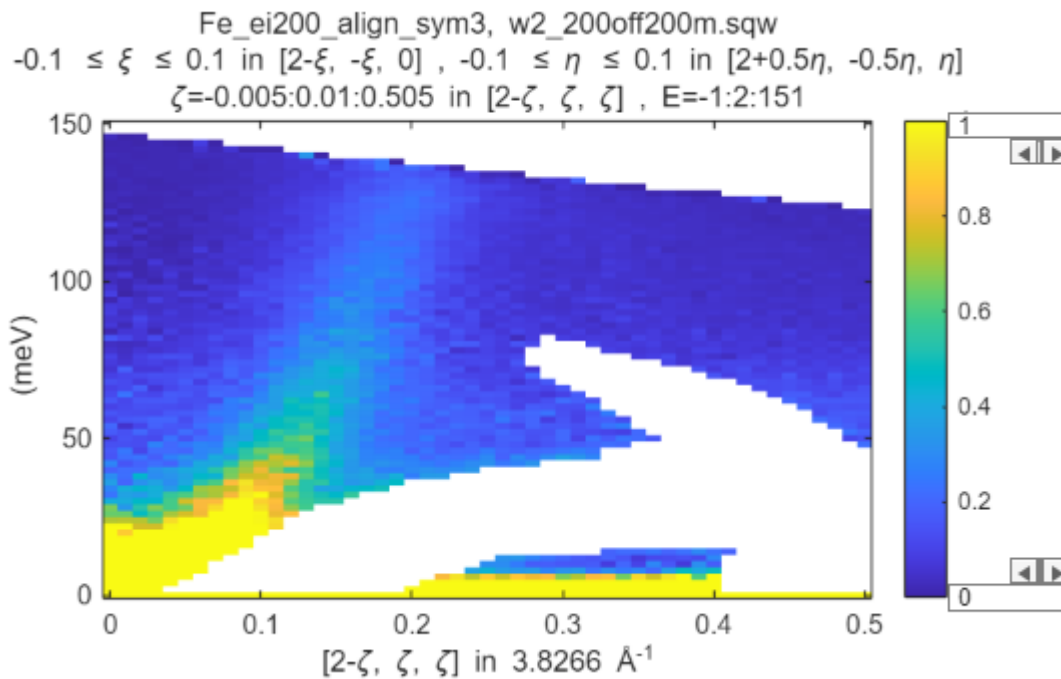
```

mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)

```

```
cuts_list{i} = cuts_list{i}.set_instrument(mi);
cuts_list{i} = cuts_list{i}.set_sample(sample);
plot(cuts_list{i}); lz 0 1; keep_figure;
```

end



```
dir_name = "GP";
```

```
dE_step = 2; %original energy transfer step data were binned to. No point in  
going finer
```

```
half_dE = 10; % half width of data binning
```

```
cut_en = sel_info.en_range;
```

```
cut_dimensions = sel_info.cut_dim;
```

```
bg_par = sel_info.bg_par;
```

```
cuts_list{end+1} = bg_par;
```

```
fbase_name = sel_info.fname;
```

```
[fit_res_rec,res_name] = fit_multicuts_along_direction(...  
cuts_list,fbase_name,dir_name,cut_dimensions,cut_en,dE_step,half_dE,true);
```

```
*****
```

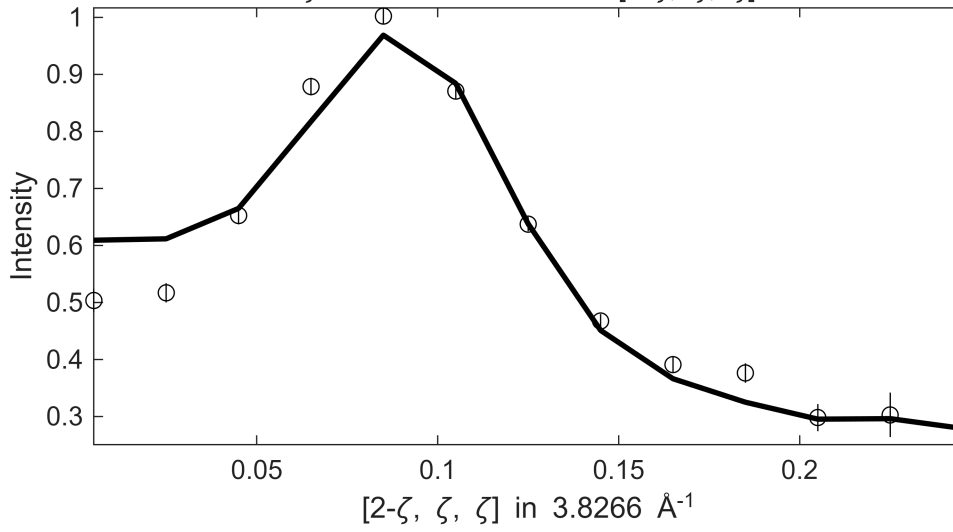
```
**** En = 30±10
```

```
*****
```

Fe_ei200_align_sym3, w2_200off200m.sqw

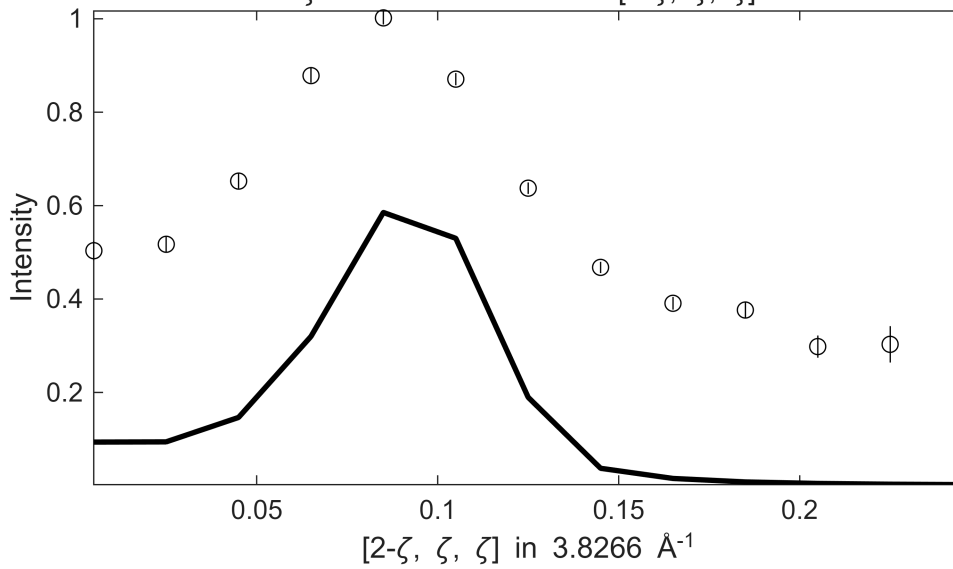
S=0.71±0.029; J0= 23±0.33; gamma=3.31188±0.112148

$0.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $20 \leq E \leq$
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off200m.sqw

$0.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $20 \leq E \leq$
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

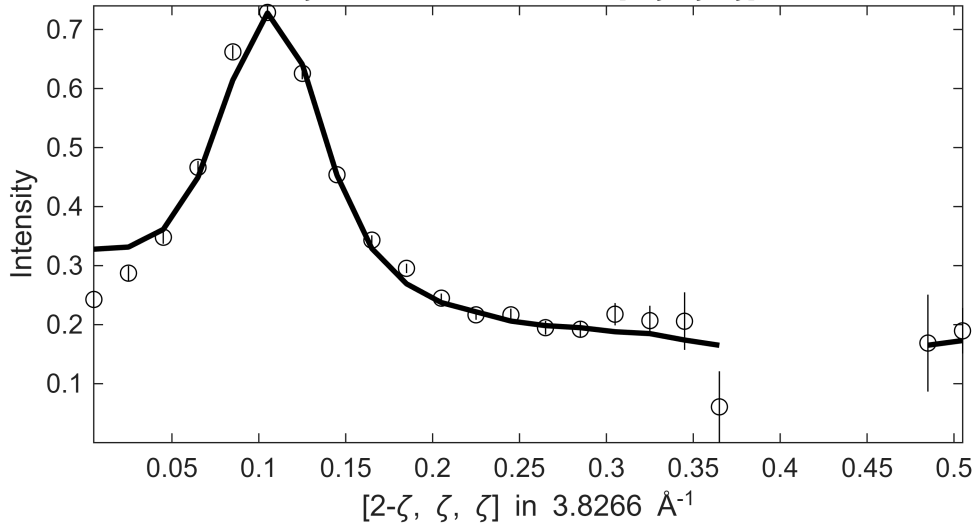


**** En = 40±10

Fe_ei200_align_sym3, w2_200off200m.sqw

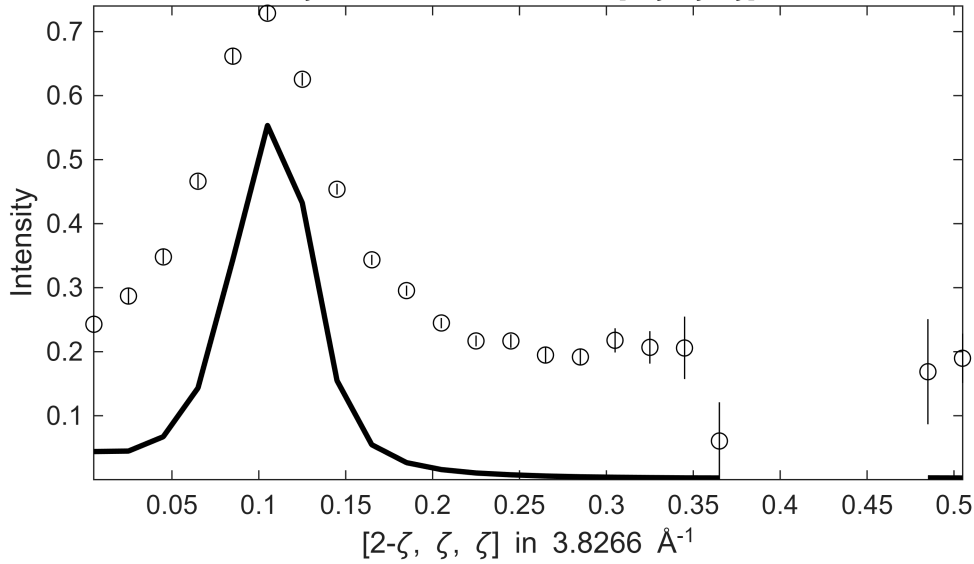
S=0.86±0.018; J0= 24± 0.2; gamma=7.54213±0.147641

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 30 ≤ E ≤
 ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]



Fe_ei200_align_sym3, w2_200off200m.sqw

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 30 ≤ E ≤
 ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]

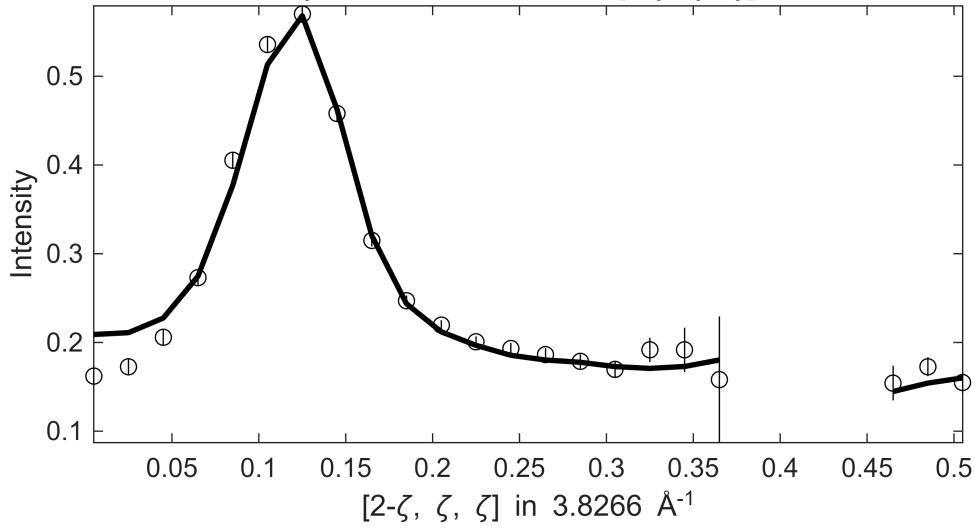


**** En = 50±10

Fe_ei200_align_sym3, w2_200off200m.sqw

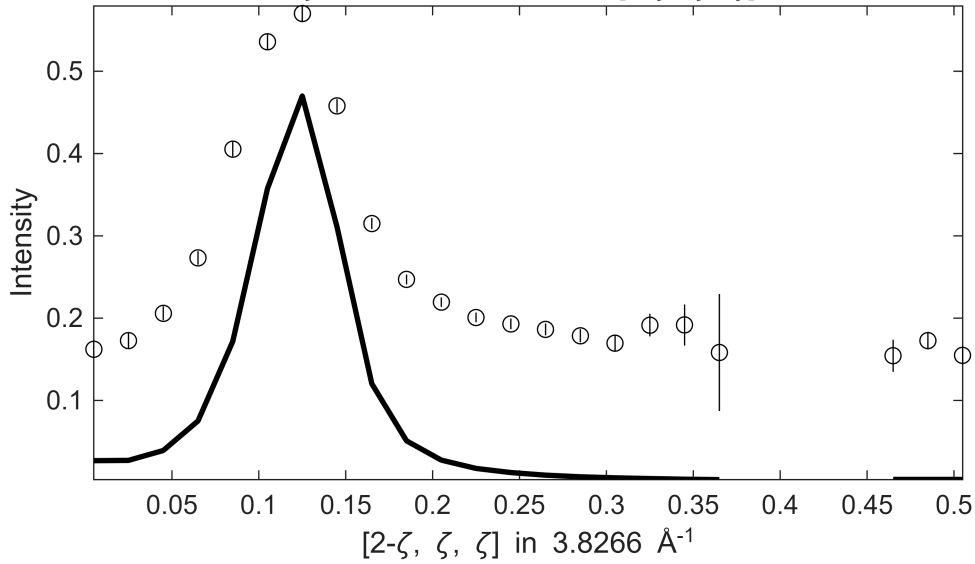
S=0.88±0.044; J0= 25±0.27; gamma=11.4193±1.11916

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 40 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]



Fe_ei200_align_sym3, w2_200off200m.sqw

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 40 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]

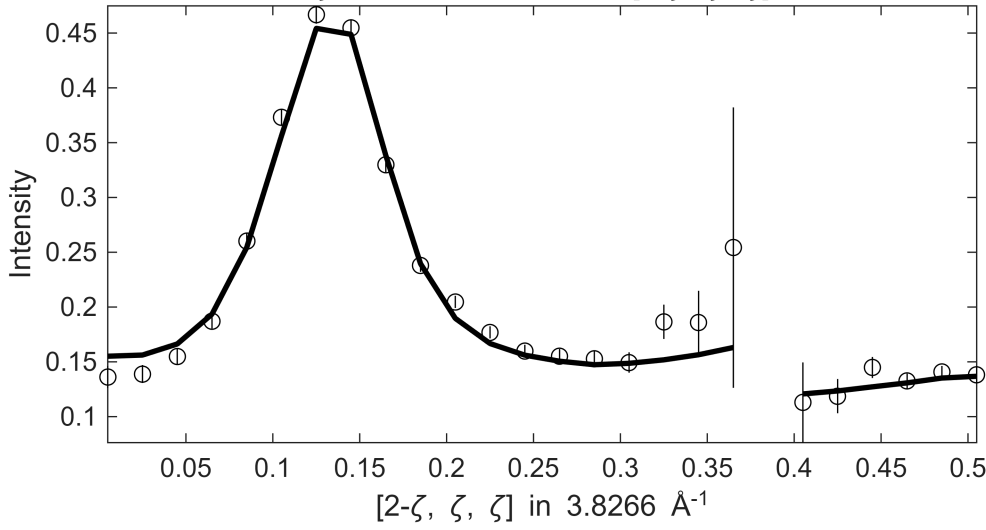


**** En = 60±10

Fe_ei200_align_sym3, w2_200off200m.sqw

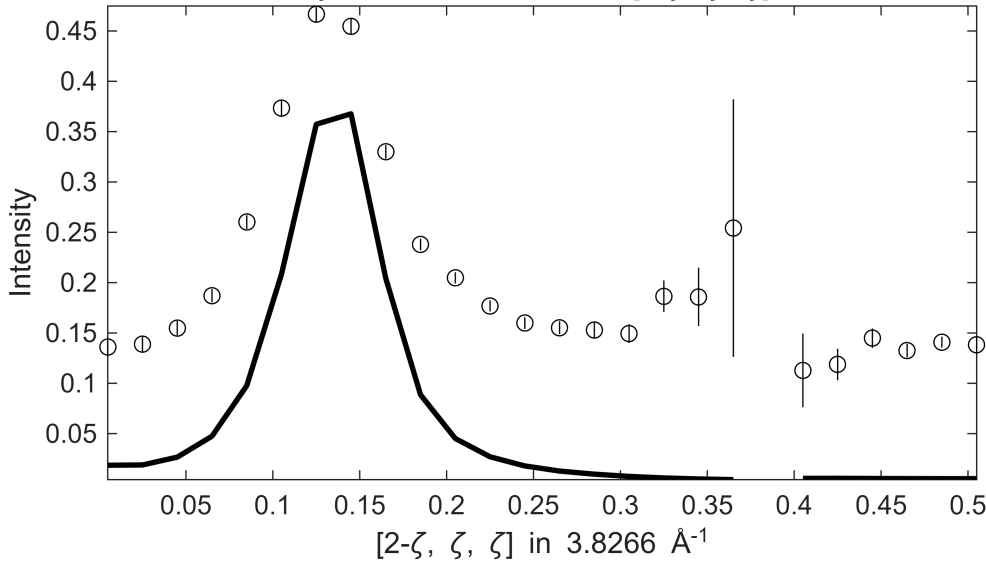
S=0.87±0.037; J0= 26±0.25; gamma=15.2357±1.15574

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 50 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]



Fe_ei200_align_sym3, w2_200off200m.sqw

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 50 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]

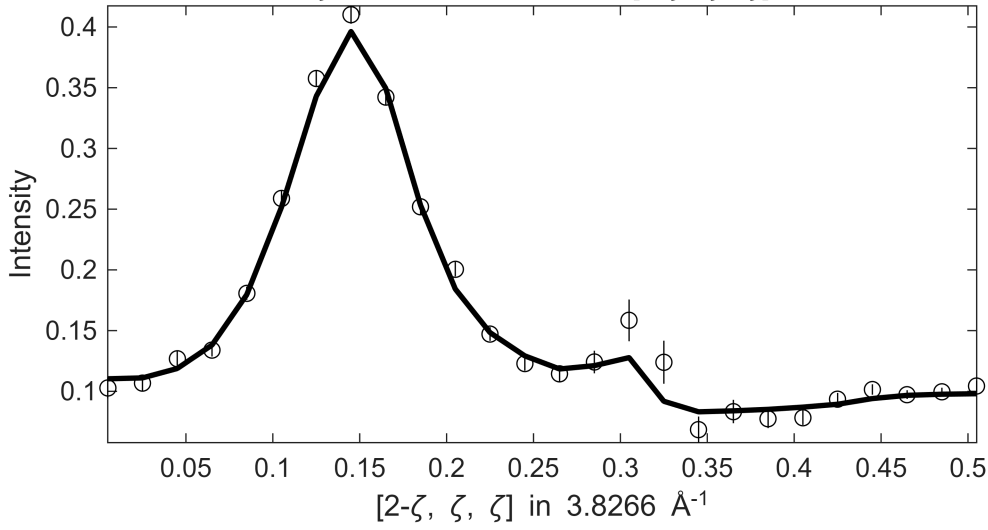


**** En = 70±10

Fe_ei200_align_sym3, w2_200off200m.sqw

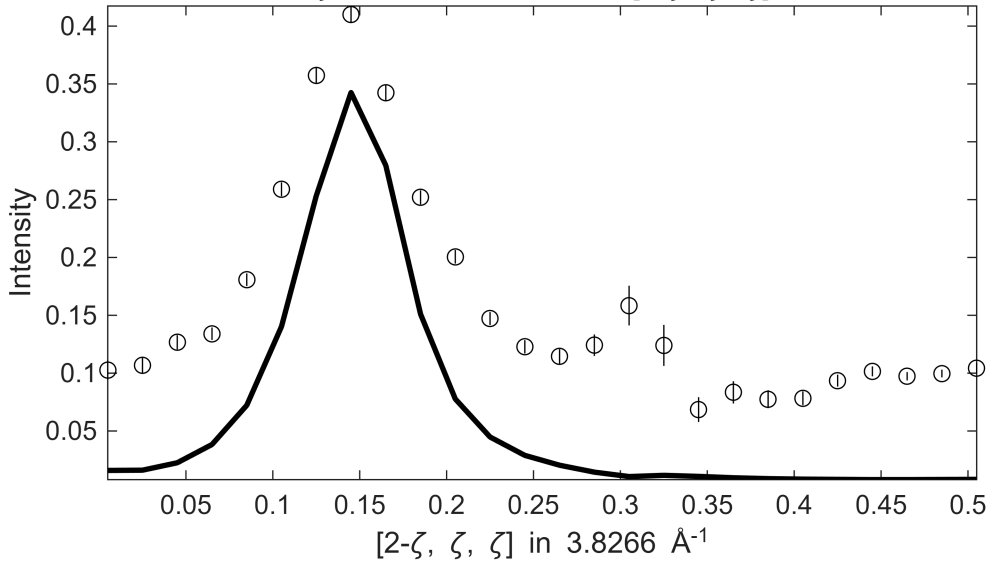
S=0.94±0.015; J0= 28±0.17; gamma=20.9144±0.300542

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 60 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]



Fe_ei200_align_sym3, w2_200off200m.sqw

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 60 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]

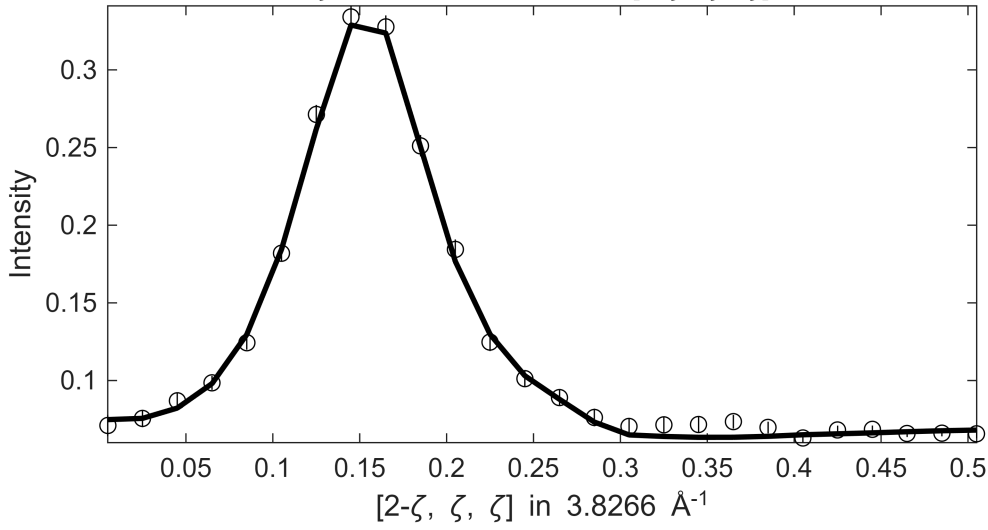


**** En = 80±10

Fe_ei200_align_sym3, w2_200off200m.sqw

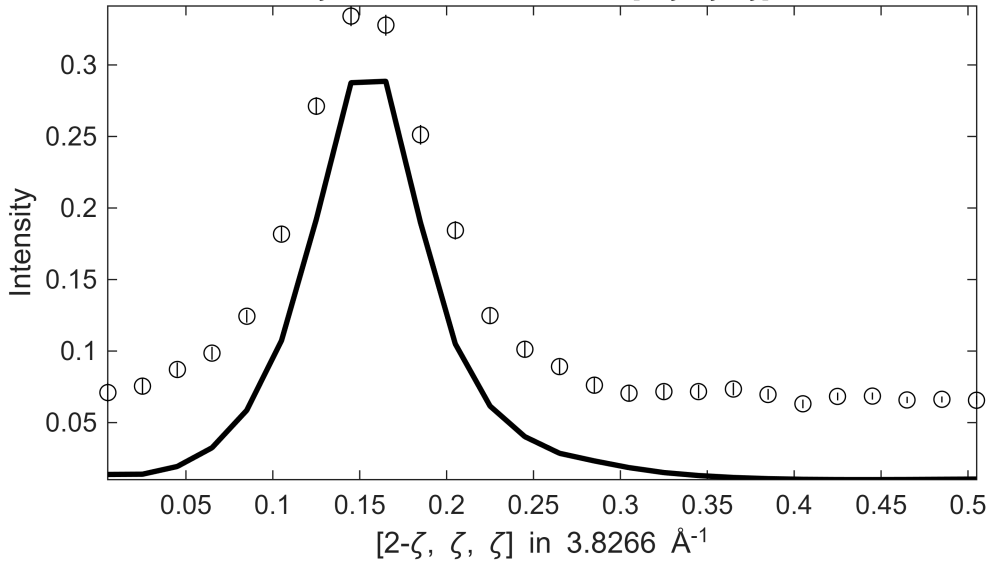
S= 1 ± 0.033 ; J0= 29 ± 0.26 ; gamma= 26.0617 ± 1.39504

$0.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $70 \leq E \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off200m.sqw

$0.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $70 \leq E \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

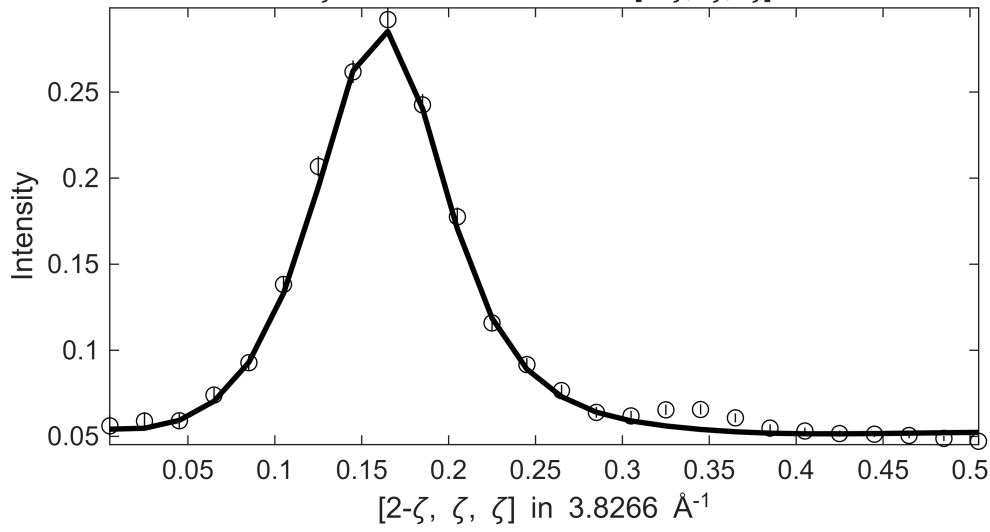


**** En = 90 ± 10

Fe_ei200_align_sym3, w2_200off200m.sqw

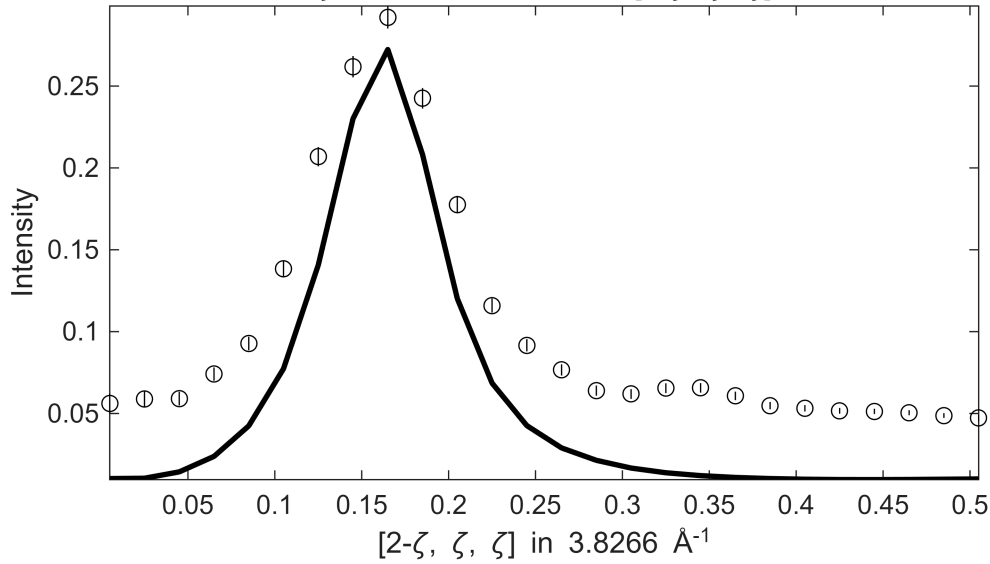
S=0.92±0.032; J0= 31±0.29; gamma=27.6858±1.56823

$0.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $80 \leq E \leq$
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



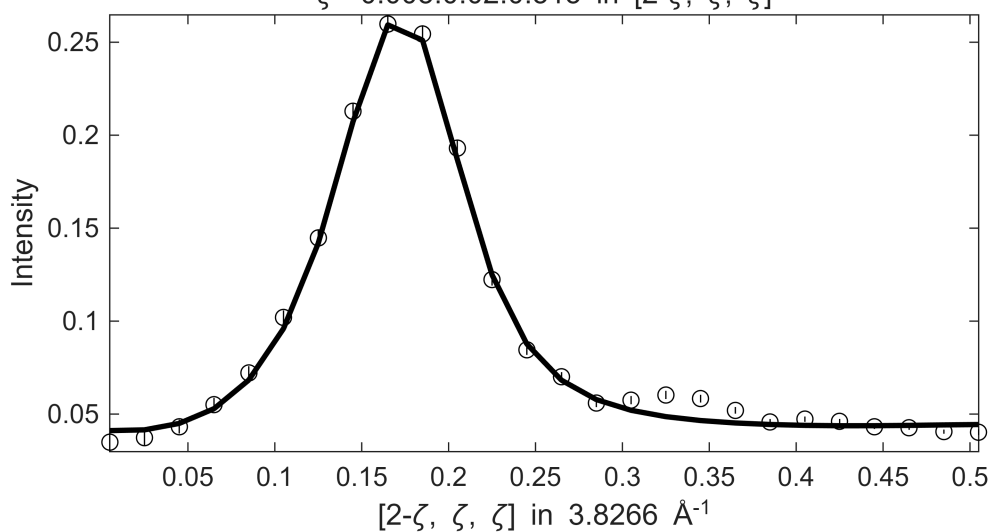
Fe_ei200_align_sym3, w2_200off200m.sqw

$0.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $80 \leq E \leq$
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

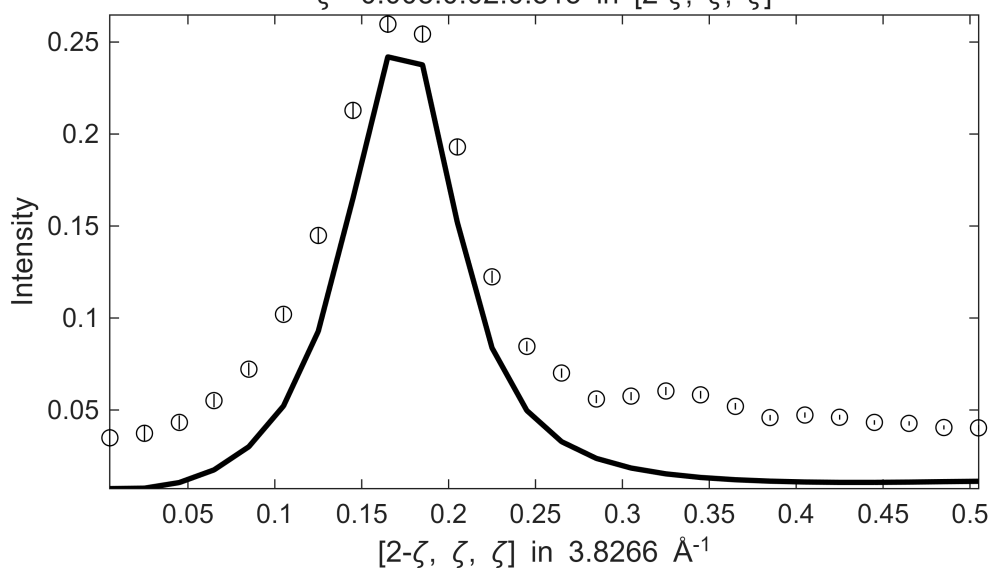


**** En = 100±10

Fe_ei200_align_sym3, w2_200off200m.sqw
 $S=0.87\pm0.028$; $J_0=32\pm0.25$; $\gamma=29.0312\pm1.51958$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $90 \leq E \leq 90$
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off200m.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $90 \leq E \leq 90$
 $\zeta=-0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

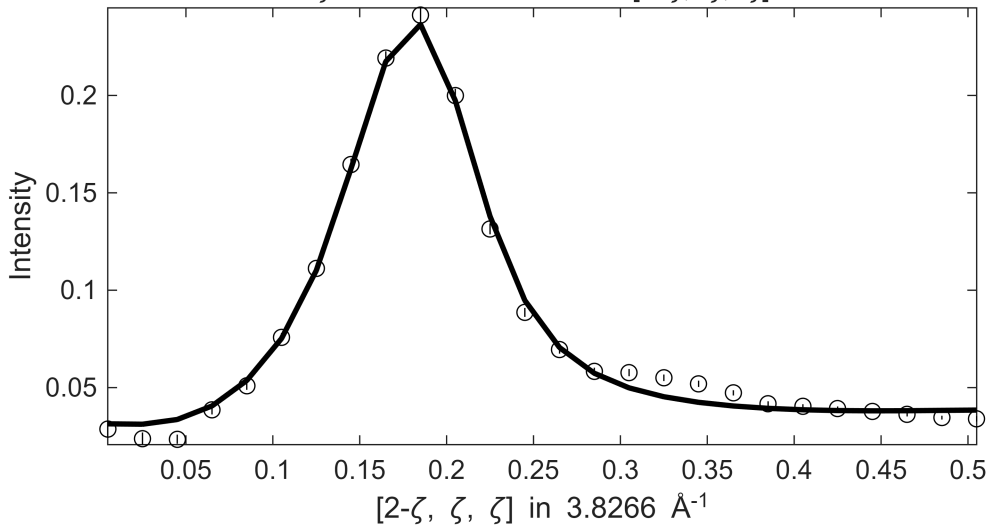


 **** $E_n = 110\pm10$

Fe_ei200_align_sym3, w2_200off200m.sqw

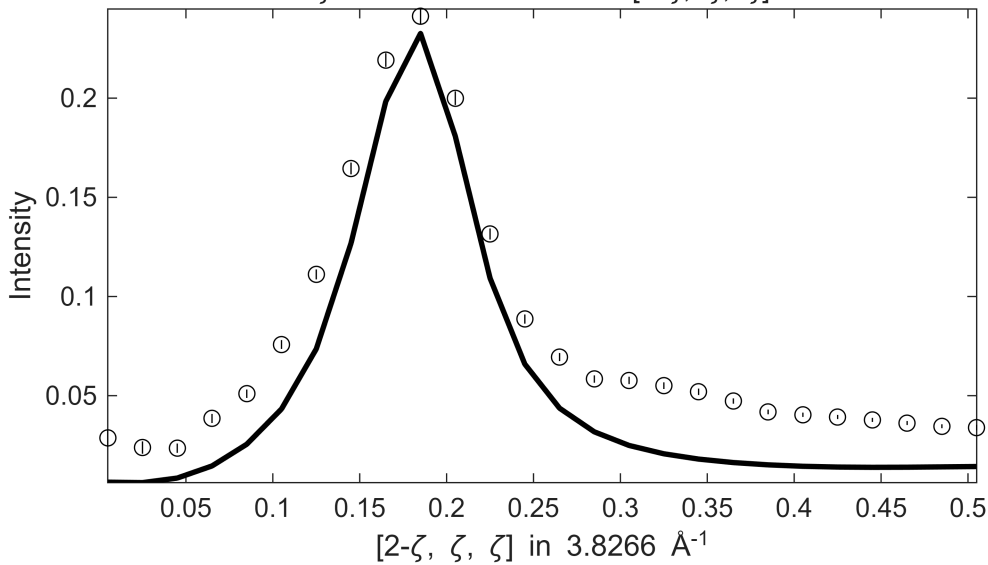
S=0.91±0.027; J0= 33±0.23; gamma=34.9196±1.5882

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 100 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]



Fe_ei200_align_sym3, w2_200off200m.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 100 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]



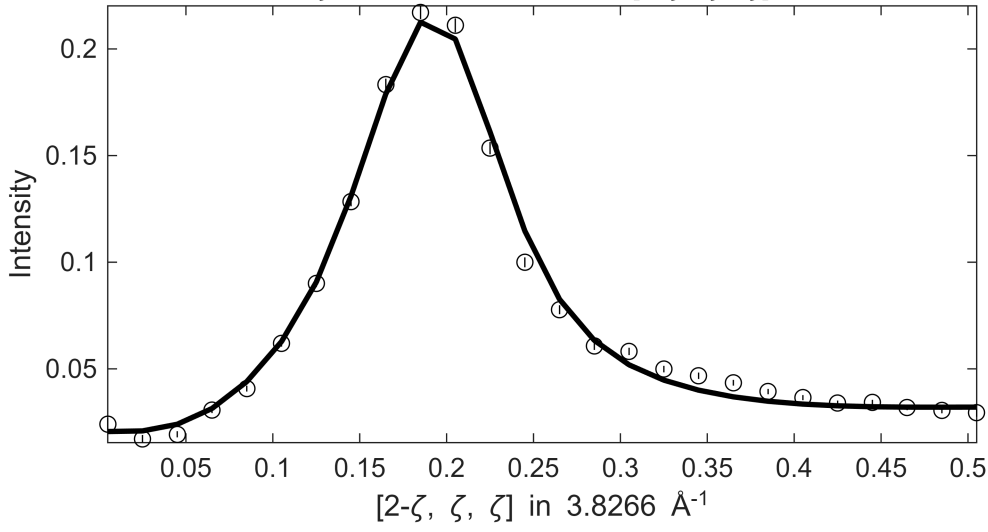
**** En = 120±10

Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.3986

Fe_ei200_align_sym3, w2_200off200m.sqw

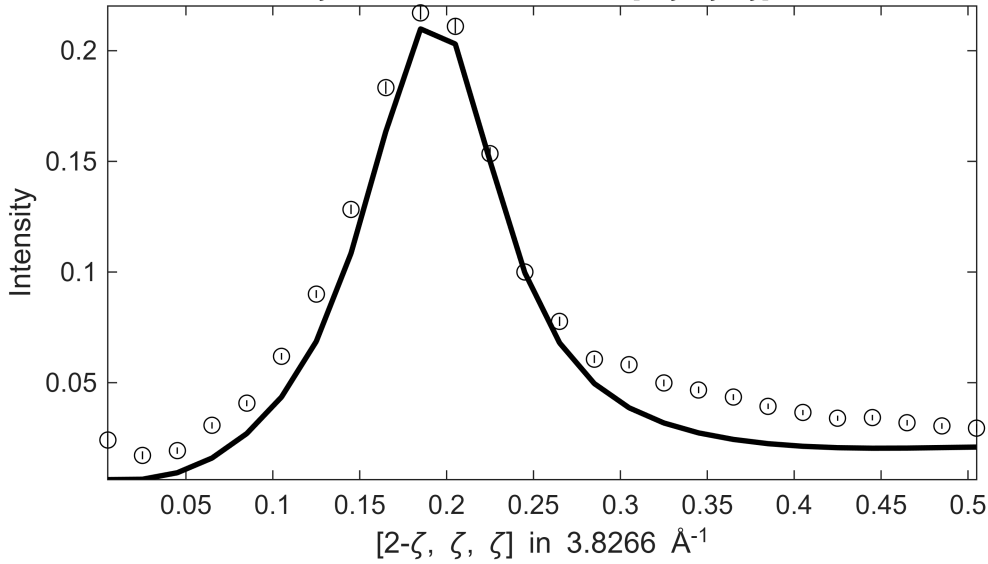
S= 1.1±0.031; J0= 34±0.24; gamma=46.8114±1.84514

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 110 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]



Fe_ei200_align_sym3, w2_200off200m.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 110 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2-ζ, ζ, ζ]



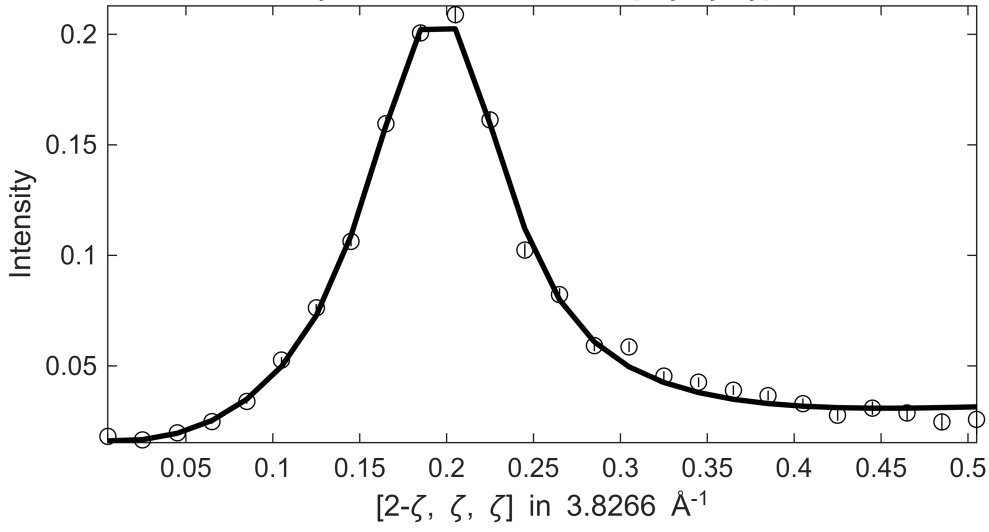
*** En = 130±10

Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.7483

Fe_ei200_align_sym3, w2_200off200m.sqw

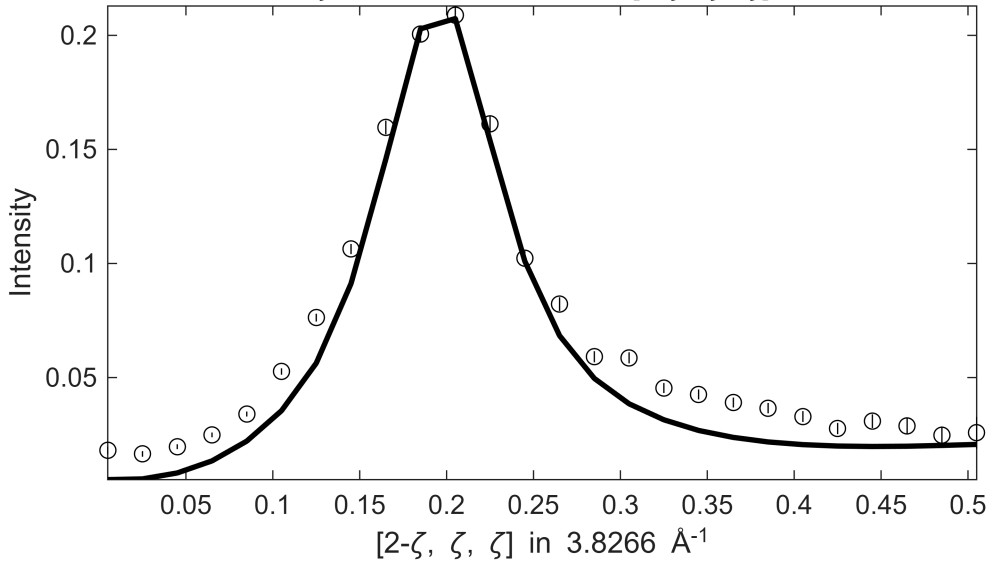
S= 1 ± 0.022 ; J0= 34 ± 0.17 ; gamma= 45.4442 ± 1.3963

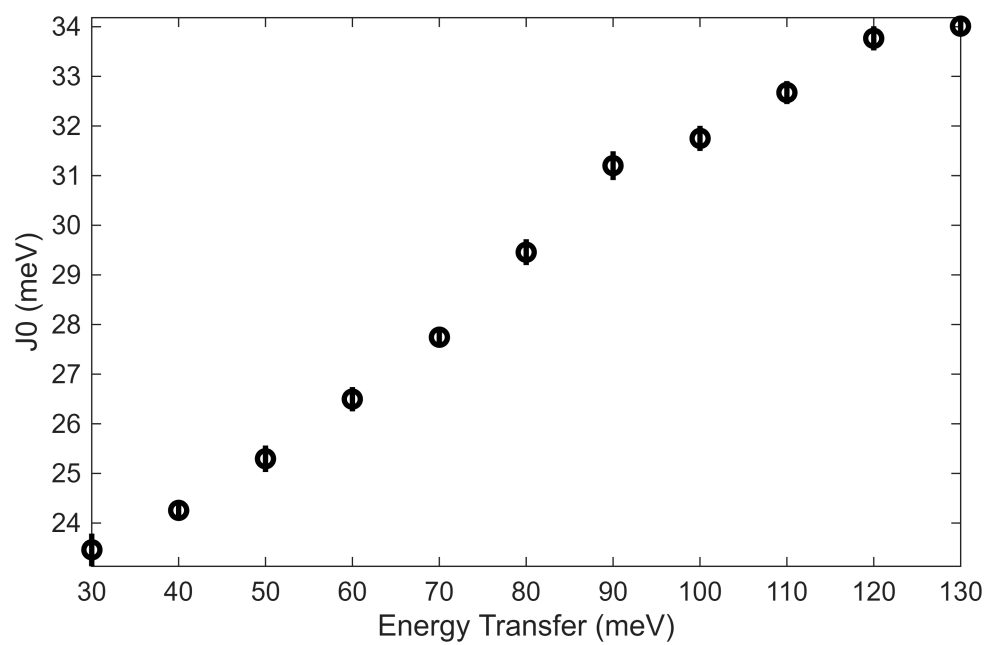
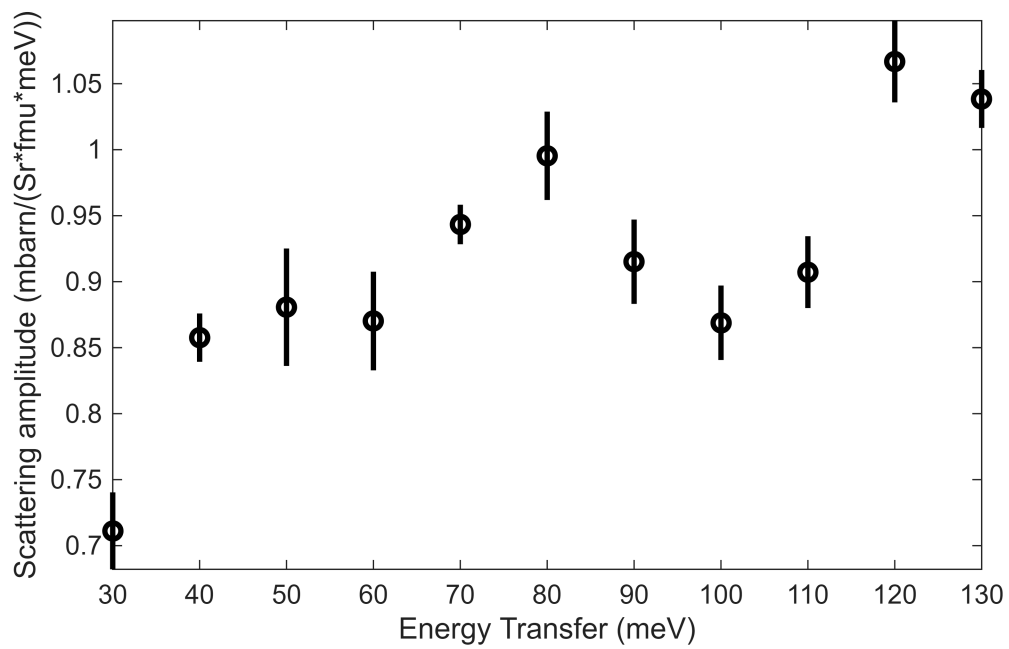
$.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $120 \leq E \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$

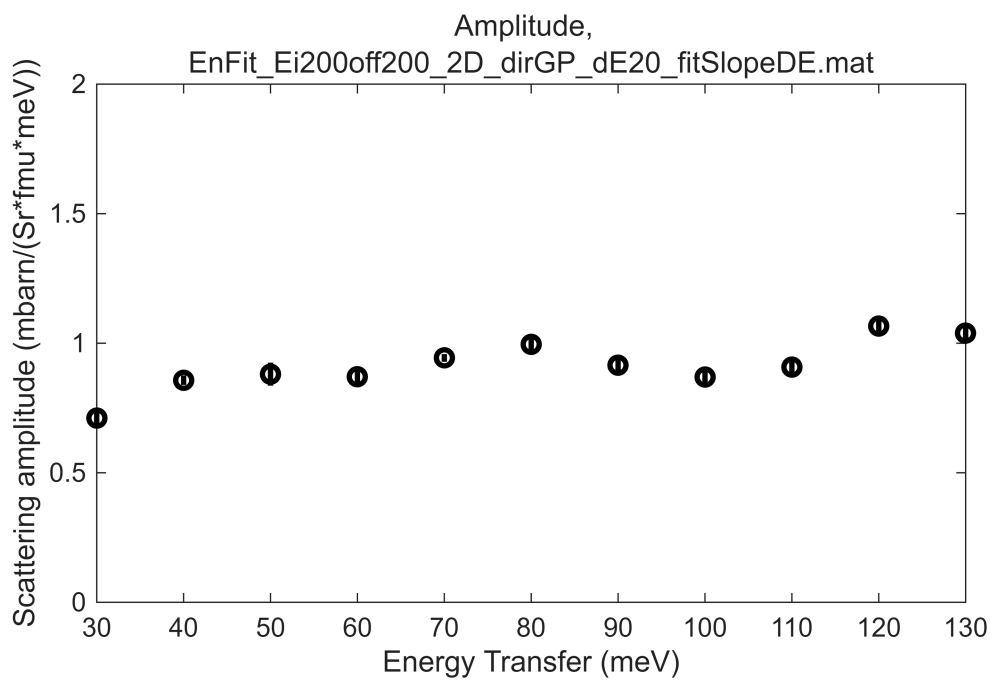
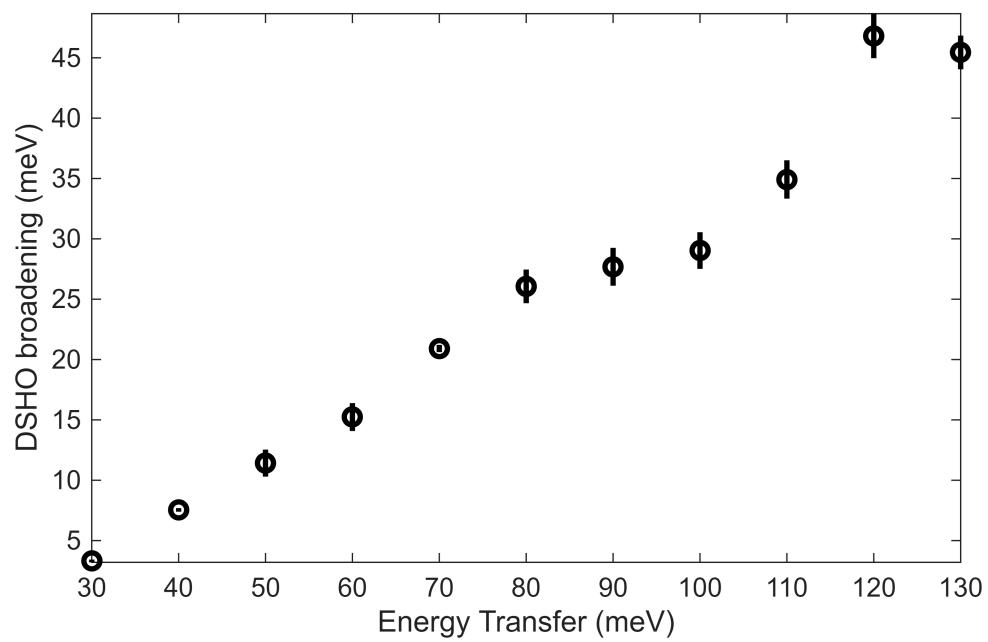


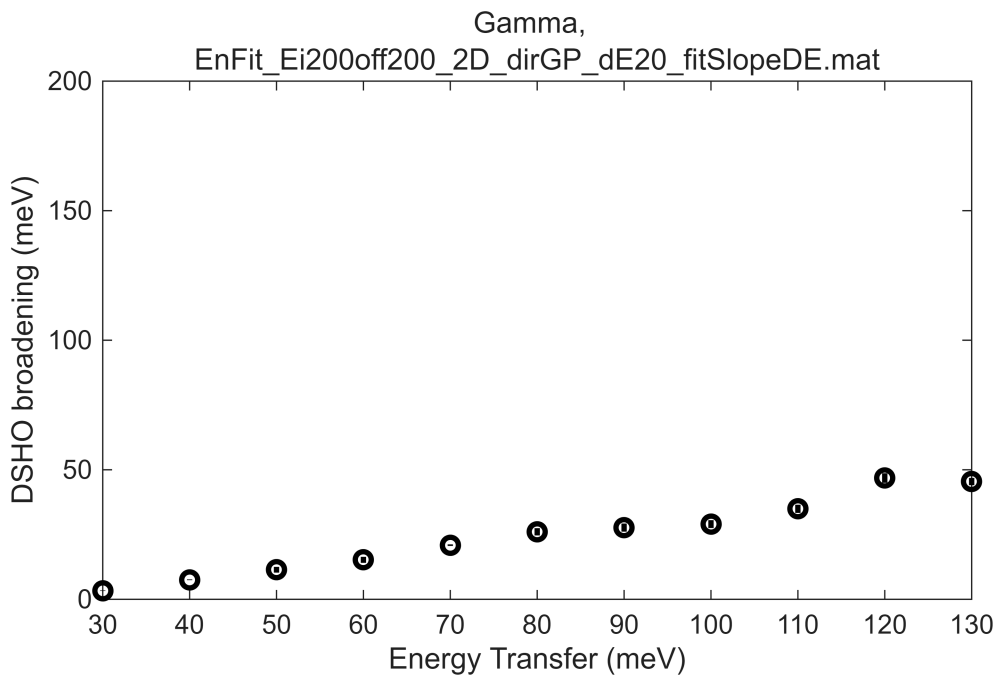
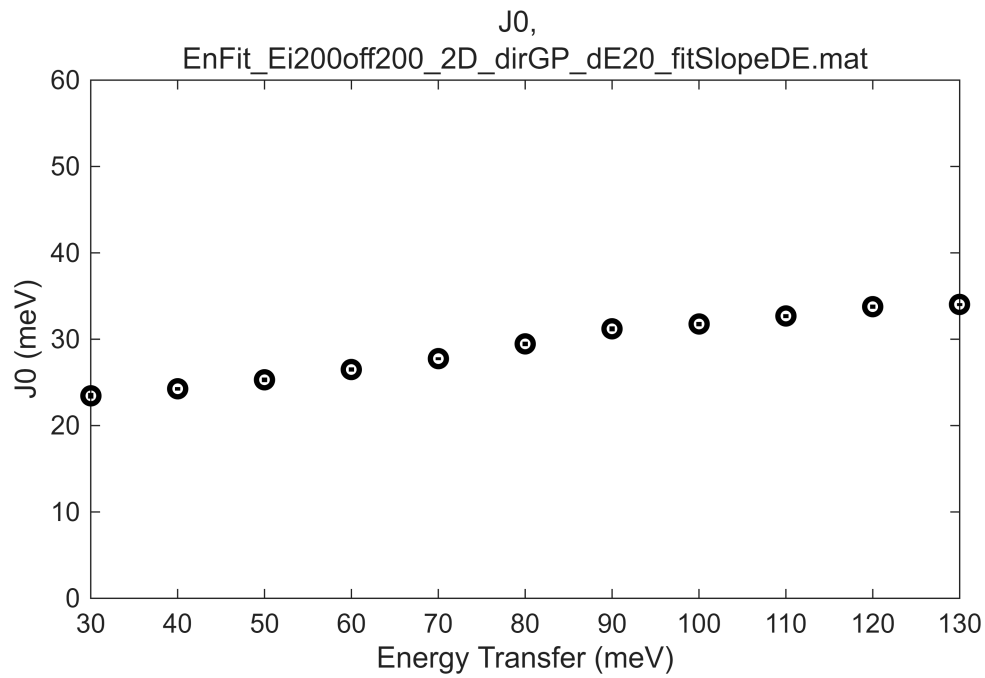
Fe_ei200_align_sym3, w2_200off200m.sqw

$.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $120 \leq E \leq$
 $\zeta = -0.005:0.02:0.515$ in $[2-\zeta, \zeta, \zeta]$





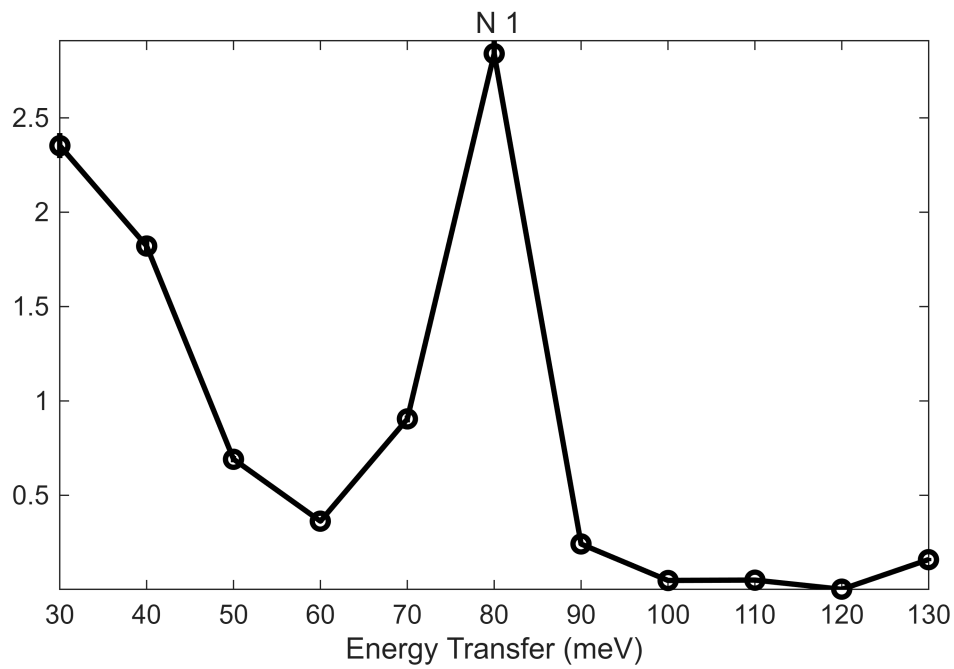




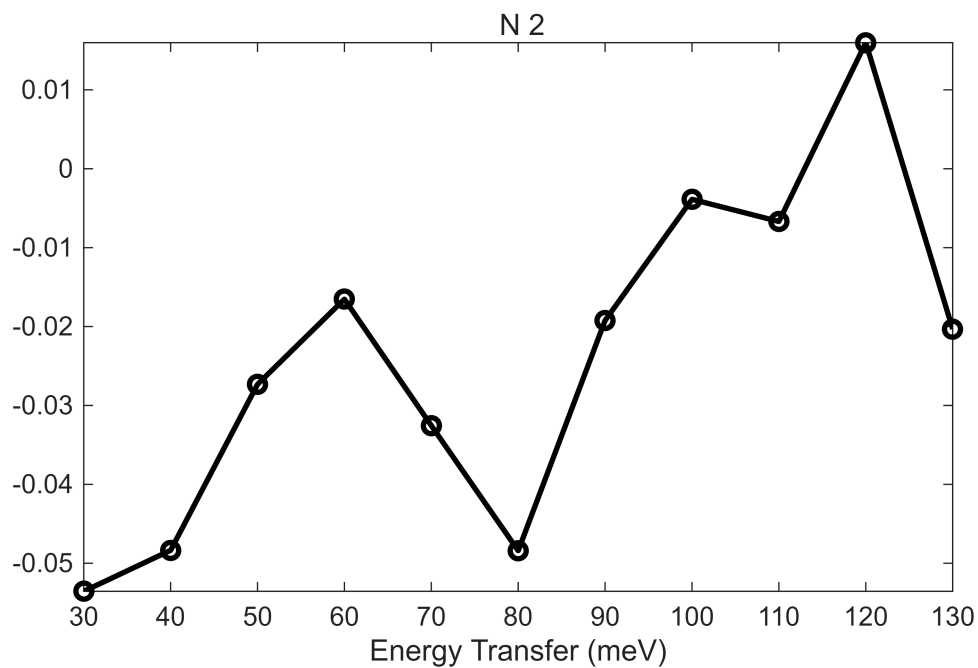
Warning: An error occurred while drawing the scene: SceneModel error in command compositeCommand:
 TypeError: Cannot read properties of undefined (reading 'deferred')
 at <https://127.0.0.1:31515/toolbox/matlab/uitools/figurelibjs/release/bundle.mwBundle.gbtfigure-lib.js?mre=https>

beep
beep
beep

```
bg_data = extract_bg_par(fit_res_rec.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);
keep_figure;
```



```
pd(bg_data(2));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);
keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure
```

```
fit_resEi200 = fit_res_rec;
save(res_name,"fit_resEi200")
```